SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID –** SIH25002
- **Problem Statement Title-**Smart Tourist Safety Monitoring & Incident Response System using AI, Geo-Fencing, and Blockchain-based Digital ID
- **Theme-** Travel and tourism
- **PS Category-** Software
- **Team ID-** 83853
- **Team Name -** Optiminds

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TRIPZAP

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Detailed Explanation of the Proposed Solution:

1.Digital Tourist ID

A **blockchain-based** ID issued at entry points **with KYC, itinerary, and emergency contacts**, valid only for the trip duration to ensure secure and private identity management.

2.Tourist Mobile App

Provides auto safety score, **geo-fencing alerts, SOS/panic button** with live location sharing, and optional real-time tracking for families or authorities.

3.AI Anomaly Detection

Monitors for sudden signal loss, **prolonged inactivity**, or route deviation, and flags missing, silent, or **distress behavior** for faster preventive rescue.

4.Police/Tourism Dashboard

Offers real-time tourist clusters and heat maps, access to IDs, alerts, last known locations, and **auto e-FIR generation** for missing/distress cases.

5.IoT Integration

Smart bands with **eSIM + GPS** send continuous health/location signals and SOS alerts, adding safety in remote or low-network regions .

6.Multilingual Support

App and portal available in **10+ Indian languages** plus English, with voice/text emergency features for elderly, disabled, or illiterate travelers.

7.Privacy & Security

Ensures end-to-end encrypted data**, tamper-proof IDs** via blockchain.

How it addresses the problem:

1. Real-time Safety & Quick Response – SOS, geo-fencing alerts, AI anomaly detection, and police dashboard ensure faster rescue and crime prevention.

2. Trust & Security – Blockchain-based tourist ID and end-to-end encryption protect identity, data, and privacy.

3. Inclusive & Smart Tourism – Multilingual support, accessibility features, and IoT wearables make the system tourist-friendly and reliable.

Innovation & uniqueness of the solution:

1. Offline SOS Alert – Unlike regular safety apps that fail without internet, this system can send a distress signal via alternative networks (SMS, local mesh, IoT wearables), ensuring tourists can call for help anytime, anywhere.

2. One-Tap Evidence Recorder – A unique feature that instantly **records audio/video proof** with one click, auto-secures it, and shares it with authorities – providing real-time evidence for faster action and legal support.

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TECHNICAL APPROACH

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TECH STACK

Frontend:

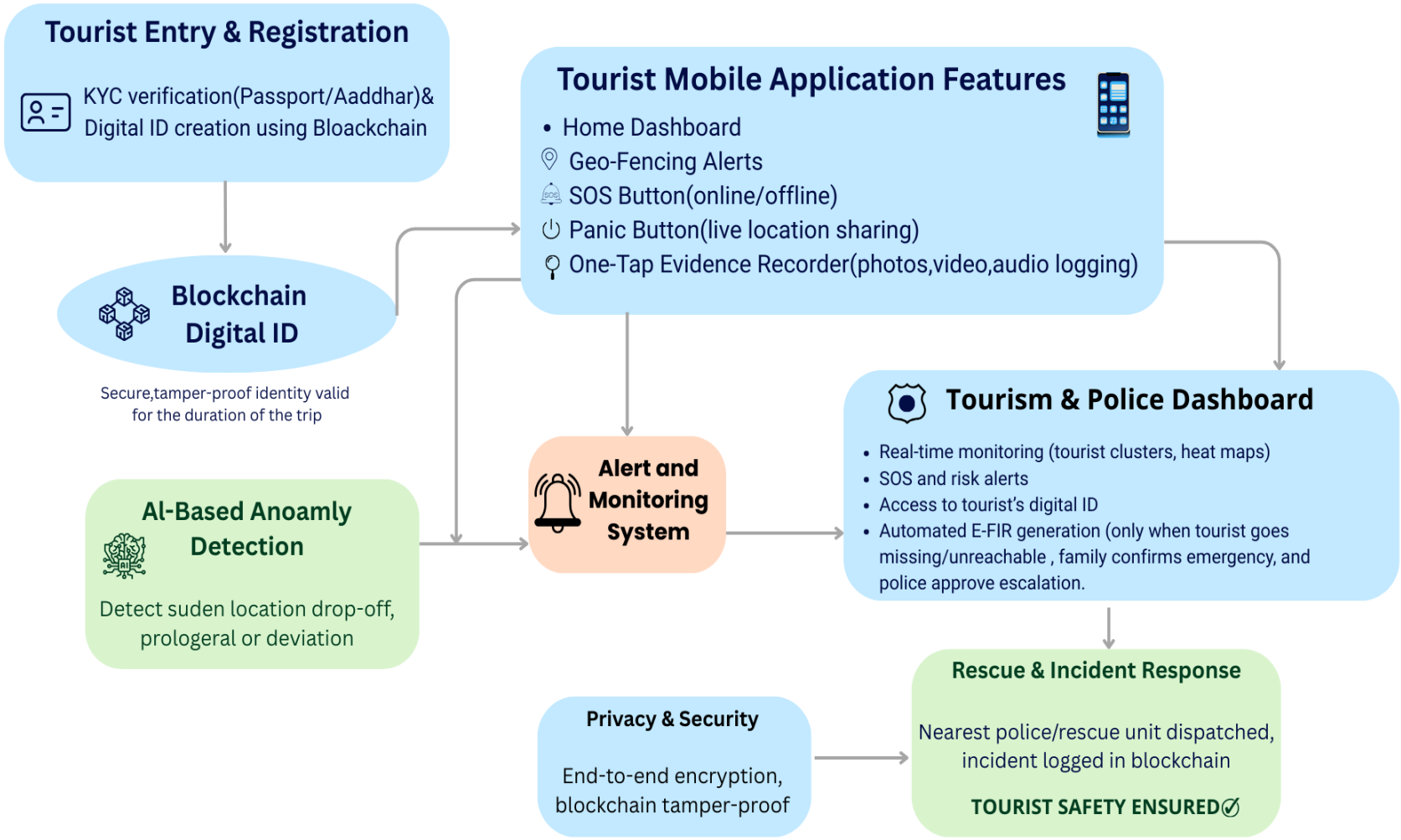
Backend:

Database:

mongoDB, SQLite, PostgreSQL

Security & blockchain:

❖ Hyperledger , JWT , IPFS & AES-256 + RSA Encryption.



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FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea:

Technical – Achievable using existing technologies (AI, blockchain, mobile apps, IoT) with moderate integration effort.

Economic – cost effective as it mainly involves infrastructure; scalable software development and deployment , requiring minimal e for large users bases without high recurring costs.

Operational – Easy for tourists (multilingual app, SOS) and practical for authorities (real-time dashboard), requiring only basic training.

CHALLENGES	SOLUTIONS
Network/Connectivity Issues: SOS and real-time tracking may fail in remote or low-network areas.	Offline SOS alerts via SMS, mesh networks, or IoT wearables ensure help is still triggered.
Data Privacy & Security Concerns: Tourists may hesitate to share location or personal info.	End-to-end encryption and blockchain-based IDs ensure secure, tamper-proof, and private data.
Integration & Adoption by Authorities: Coordinating police and tourism boards can be complex.	Unified dashboard, staff training, and automated alert system for seamless monitoring and faster response.

IMPACT AND BENEFITS

Potential Impact On Tourist:

1. **Improved Safety & Security** – Real-time alerts, SOS, and AI monitoring reduce risks of accidents, crime, or getting lost.

2. **Faster Emergency Assistance** – Immediate response from police and family through live tracking and offline SOS.

3. **Peace of Mind & Confidence** – Secure digital ID and privacy measures make tourists feel safe while exploring.

4. **Inclusive Experience** – Multilingual support and accessibility features ensure all tourists, including elderly and differently-abled, are covered.

Benefits Of The Solution:

1. **Social**: Improves tourist safety and confidence, ensures inclusivity with multilingual and accessibility features, and strengthens trust between tourists and authorities.

2. **Economic**: Increases tourism revenue by attracting more visitors, reduces costs of emergency response and crime management, and supports smarter planning for tourism infrastructure.

3. **Environmental**: Helps control tourist flow in sensitive areas, reduces overcrowding, and promotes responsible and sustainable tourism practices.

RESEARCH AND REFERENCES

- https://arxiv.org/abs/2401.03530
- https://arxiv.org/abs/2402.11231
- https://www.mdpi.com/2073-431X/14/7/247
- https://www.mdpi.com/2076-3417/15/15/8330